**COMP 2087: Programming for Beginners II**

Assignment #3

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For each question, write a program **with comments** to perform the required tasks.
(Q1 = **30** points; Q2 = **37** points; Q3 = **28** points. TOTAL = **95** points)

1.

a. Write a function *fib1()* using iteration, that takes an int ***max*** as input, and calculates and prints all the Fibonacci numbers below ***max***. The function also returns a value indicating how many numbers in the Fibonacci sequence have a value less than ***max***.
b. Write a function *fib2()* using recursion, that takes an int ***n*** as input, and calculates the $n^{th}$ Fibonacci number.
c. Randomly generate an int ***max*** between 25 and 250. Call *fib1(**max**)* and print the value returned by *fib1()* with a suitable message.
d. Ask the user to input a number ***n*** between 5 and 16 You can assume the user will enter a valid input in the proper range. Use the function *fib2()* to generate and print the first ***n*** numbers in the Fibonacci sequence.

Sample outputs (for parts c and d respectively):

```
"C:\Program Files (x86)\Python312\pyt
max = 180
Fibonacci seq using fib1()
0
1
1
2
3
5
8
13
21
34
55
89
144
There are  13 fibonacci nos. <  180
```

```
Enter an int between 5 and 16: 14
Fibonacci seq using fib2()
0
1
1
2
3
5
8
13
21
34
55
89
144
233
```

2. In a file *student.py* create class definitions as indicated below.

a. Define a class called **Student** with instance variables for the `studentID` (int), student's `name`, `gpa` ( a number between 0 and 100, displayed as float rounded to 1 decimal place) and `courses` ( the number courses used for `gpa` calculation) and. When creating an instance of the class, default values of `'unknown'` 0.0

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and 0 can be used for `name, gpa` and `num_courses` respectively; but the `studentID` must be specified. All instance variables should be considered 'private'

b. Write the following methods for this class:
   i. `__init__()` # create and initialize new object with given parameters
   ii. `getCourses()` # returns current value for number of courses
   iii. `add_mark()` # user specifies a course mark (int between 0 and 100) to be included in the `gpa` calculation. **Note**: value of `courses` must be incremented as well when a mark is added, but cannot be more than 40. If the `gpa` is already based on 40 courses, a suitable message should be displayed to the user. The function returns the `gpa` for the student (after any update).
   iv. `del_mark()` # user specifies a course mark (int between 0 and 100) to be excluded from the `gpa` calculation. **Note**: value of `courses` must be decremented as well, but cannot be less than 0. If the current `gpa` is based on 0 courses, a suitable message should be displayed to the user.
   v. `__str__()` # shows info in format: `studentID(name): gpa`.

c. Define a *subclass* of **Student** called **GradStudent**, which has another instance variable called `_extra`, that specifies the number of additional courses (above 40) that the student is allowed to take. The default value is 10.

d. Override the `add_mark()` method of the parent class. If current value of courses is ***c*** , the new *add_mark* method does the following:
   i. If ***c*** < ***40*** then new balance is updated regularly.
   ii. Otherwise, if ***c*** < 40 + ***_extra*** then the new mark is used calculate the updated `gpa` and a warning message is printed indicating how many extra courses the student has added.
   iii. Otherwise, gpa update is not allowed, and a suitable message is printed.

3. Import *student.py* then create 5 objects using the information given below. If a value is not specified, use the default value.

| Type | studentID | name | gpa | courses | extra |
|---|---|---|---|---|---|
| Student | 1111 | Alice | 85.7 | 35 | n/a |
| Student | 2222 | Bob | 90 | | n/a |
| Student | 3333 | Cathy | 63 | 40 | n/a |
| GradStudent | 4444 | Charlie | 55.8 | 40 | 5 |
| GradStudent | 5555 | Daisy | 74 | 40 | |

   a. Create a list called ***studentList*** containing the 5 objects that were created.
   b. Print the information for each student in StudentList
   c. Try to remove a mark of 74 from gpa calculation of and print the new gpa with a suitable message for .

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i. Student with ID 1111
ii. Student with ID 2222

d. For each account in ***studentList***, print the account information then try to add a mark of 65 for gpa calculation.

Sample output:

```
"C:\Program Files (x86)\Python312\python.exe" "C:\User
***Part b: Info for all students***
1111(Alice): 85.7
2222(Bob): 0.0
3333(Cathy): 63.0
4444(Charlie): 55.8
5555(Daisy): 74.0
***Part c: Example of del_mark()***
Alice  new gpa:  86.0
There are no marks to delete for Bob
Bob  new gpa:  0.0
***Part d: Example of add_mark() for all students***
1111(Alice): 85.4
2222(Bob): 65.0
Cathy has exceeded 40 courses, cannot add more marks!
3333(Cathy): 63.0
Warning for Charlie : 3  extra course(s) added.
4444(Charlie): 56.0
Warning for Daisy : 1  extra course(s) added.
5555(Daisy): 73.8
```